In this brief statement, I will outline my activity versus expectation for each faculty pillar of teaching, research, and service.

Teaching Excellence

ERAU-WW COAS expects Associate Professors to demonstrate excellence in teaching. This is described in the Guidelines and Criteria document as follows:

> COAS embraces a philosophy of teaching effectiveness that emphasizes knowledge of subject matter, enthusiasm in the classroom, and continuous improvement through the incorporation of new teaching methods and materials. Faculty are expected to effectively communicate with students, facilitate learning in the classroom, and promote attainment of Student Learning Outcomes and General Education Competencies. These activities should be fostered through regular participation in pedagogical development opportunities and appropriate course maintenance and redevelopments.

I hold exceptional subject matter expertise in the courses I teach. As part of my Ph.D. in Mathematics and Statistics, I completed 33 graduate-level credit hours of mathematics courses [**CV, Notable Coursework, pg 13**], including Ph.D. qualifying exams in Matrix Analysis and Abstract Algebra. This high-level coursework allows me to contextualize undergraduate mathematics in a way that allows for productive abstraction and transfer to related topics of study. Abstract Algebra provides foundational mathematics for cryptography, programming, and physics while Matrix Analysis is critical to data analysis, machine learning, and 3D movement of objects (like a flying plane). For example, a common task in a Precalculus course is to solve a System of Equations. I can contextualize this task within aeronautics as 2-dimensional balance of forces (gravity, drag, thrust, etc), then use technology to show students how these calculations might be made by a computer for the "real" 3-dimensional version. For students more interested in machine learning, this same task of solving a System of Equations can be contextualized as computer image recognition. This allows me to spark student curiosity that can drive their engagement with mathematics.

As I learned completing 15 graduate-level credit hours focused specifically on teaching college mathematics [**CV, Notable Coursework, pg 13**], teaching well is more than just subject knowledge expertise. Research in Collegiate Mathematics (and Education in general) suggests subject knowledge acts in unison with pedagogical and social knowledge to deliver educational experiences. Thus it is critical to not only know the subject well, but also to know how to communicate these ideas at the level the student understands the concept. My continued research on how students develop mathematical ideas fuels my continued improvement as an instructor.

The union of my subject matter and pedagogical expertise allows me to provide exceptional teaching in ERAU-WW courses. This is evidenced by my overwhelmingly positive student evaluations [**Student Course Evaluation Narrative**] that explicitly mention how my knowledge and enthusiasm create an effective learning environment. My research also affords me the opportunity to constantly incorporate new methods and materials. I have been part of three funded external grants from the National Science Foundation (NSF) Improving Undergraduate STEM Education (IUSE) program, received a mini-grant from a separate NSF IUSE grant, received an ERAU Faculty Innovative Research in Science and Technology (FIRST) grant, and received two ERAU-WW Faculty SEED grants that target improving specific Embry-Riddle courses [**CV, External/Internal Research Funding Experience, pgs 1-2**]. These research projects provide tangible improvements to student learning at Embry-Riddle, such as the Doenet mini-grant to develop autonomous discovery-based learning online [**CV, External Research Funding Experience, 8th item**]. I have also incorporated traditional active learning when teaching EagleVision sections, as noted by Zackery Reed's peer observation of my October 2022 Math 111 section. From his review:

> Dr. Chamberlain demonstrates a productive balance of direct instruction (i.e., lecture) and active learning activities. This suggests an intentional awareness of student attention span needs, and more importantly an implementation of currently known best practices for mathematics pedagogy.

Curriculum Development

I currently run two programs within the Department of Mathematics, Science, and Technology: the Minor in Applied Data Science and the Bachelors in Data Science. Both have seen early success in their enrollment numbers, as listed in the chart below. These programs leverage the world's interest in data science with the popularization of LLMs and other Artificial Intelligence software.

	Minor DSCI	Bachelor DSCI
August 2023	2	NA
January 2024	2	NA
August 2024	4	NA
January 2025	11	NA
August 2025	26	19
January 2026	42	34
August 2026	72	54

Given my extensive experience with course development during graduate school and while at the University of Florida [**CV, Teaching Experience, pgs 10-11**], I have completed several course developments, redevelopments, and updates. My developments have been widely praised by Instructional Design and Development (IDD) [**IDD Support Letter**], faculty reviewers, and students. While my [**Summary of Teaching Effectiveness**] details all significant development efforts, I present the most impactful development below.

MATH 502 - Boundary Value Problems: I was asked to teach MATH 502 in August 2024 due to sustained student complaints and notices that students were dropping out of the Master's in Aerospace Engineering. Students were generally frustrated with the course structure and overly challenged by the material itself. I made several small changes to the course to address student complaints, including: (a) shifting all content back a week and creating a review of Ordinary Differential Equations for Module 1, (b) creating reflective discussion posts that encouraged students to explicitly link how the course topics related to one another, and (c) replacing synchronous oral exams with asynchronous oral exams. Creating the Module 1 review of Ordinary Differential Equations was enough to reduce student work on the proceeding Modules from an average of 30 hours per week to about 15 hours per week, putting it in line with expectations for a graduate-level course workload. Students generally praised the changes made to the course:

> Great professor for the class, especially with this course because MAN this course was the hardest class I've had in years. Dr. Chamberlain should change ALL the videos to his style because HE can teach this like its NOTHING, such a great professor! (Aug 2024)
>
> The professor was really good at explaining the content when I had questions. I was impressed by how open and adaptive he was with feedback from the class. He recognized that somethings weren't working and changed quickly to ensure we learned in a better way. (Jan 2025)
>
> Out of the two courses I've taken this takes the cake. The lectures were well-structured, and Dr. Chamberlain's teaching style struck a great balance between theory and practical problem solving. I especially appreciated the timely, specific feedback on assignments. The course also made room for growth; policies and discussions encouraged iteration and improvement rather than one-and-done grading. The materials (notes, example codes, and references) were organized and directly useful for quizzes and projects. (Aug 2025)

Scholarly and Professional Activity

My research agenda focuses on applying theoretical learning trajectories and machine learning algorithms to analyze, anticipate, and describe student thinking, especially in the mathematical sciences. My scholarly activity, compared to total expectations outlined in the COAS ERAU-WW Academic Guidelines and Criteria for Tenure and Promotion to Associate Professor, is provided in Figure 1.

	Peer-Reviewed Publications			Scholarly Presentations					Funding Proposals			
	Book Chapters	Journal Articles	Proceedings Papers	International Conference	National Conference	Regional Conference	Local Conference	Invited Talks	Funded External	Unfunded External	Funded Internal	Unfunded Internal
Total	**1**	**7**	**9**	**3**	**19**	**1**	**4**	**6**	**4**	**6**	**4**	**0**
Year 1: AY21-22		1	2		4	1		1	1		1	
Year 2: AY22-23		2	1		1		1	2	1		1	
Year 3: AY23-24			1		4			2	1	1	1	
Year 4: AY24-25	1	1	4	1	5		2	1		2	1	
Year 5: AY25-26		3	1	2	2		1		1	3		
Year 6: AY26-27					3							
Actual Sum Total	17			23			4	6	14			
Expectation	5			5			NA	NA	1			

Figure 1: Summary table of activity for AY1-6

Of the peer-reviewed publications, 10 of 17 were first-author publications and 3 of 17 included student researchers. Publications were made in high-quality venues focused on education. The book chapter was published in *Teaching and Learning Mathematics Online 2e*, the second edition of a series focused on presenting meaningful and practical solutions for teaching mathematics online. Proceedings papers were primarily published in the proceedings of the annual conferences on Research in Undergraduate Mathematics (6 of 9) and American Society for Engineering Education (2 of 9). Journal articles went to high-quality journals such as *Journal of Mathematical Behavior*, *Active Learning in Higher Education*, and *Problems, Resources, and Issues in Mathematics Undergraduate Studies*.

All proceedings papers had an associated international (1) or national (8) presentation. The other 2 international presentations were associated with an internal grant (FIRST) and included 1 graduate and 2 undergraduate student researchers. The graduate student decided to continue with their PhD after finishing their involvement with the project, stating in their exit interview that working on the project helped them learn how to structure code for a large-scale research project and formalize it for publication. Other national presentaions were primarily at the annual Mathematical Association of America MathFest conference (5 of 11).

In terms of external grants, I have been a co-PI on (3) funded NSF grants and PI on (1) subaward attached to NSF grant while being PI (3) and co-PI (3) on unfunded external grant proposals. The chart does not include a resubmission of an unfunded but competitively-rated grant in October 2026 nor does it include a projected resubmission of another unfunded but competitively-rated grant in January 2027. I have had stellar success with internal funding: I have been PI (2) and co-PI (2) on funded college-level (3) and university-level (1) internal grants. My summary of scholarly activity provides a short description of how my active research projects contribute to my overarching research agenda on utilizing machine learning and hypothetical learning trajectories to model student thinking and learning [**Summary of Scholarly Activity**].

Service

Finally, ERAU-WW COAS expects Associate Professors to provide service both to ERAU-WW and beyond. Specifically, the Guidelines and Criteria document call for:

> Appropriate service to their department, college, campus, university, and professional community. Service commitments for a candidate will be determined annually in collaboration with the Department Chair. Candidates' portfolio should include evidence of varied, sustained and, typically, progressive levels of service.

I have had significant service to the department and college, as summarized in my [**Summary of Service**]. Of these, one of my most impactful services was stepping in as Associate Department Chair after former department chair Karen Keene suddenly passed in February 2023. In my annual evaluation, interim department chair Beverly Wood stated *"He has been a key ingredient to the recovery of MST following the sudden death of Karen. Not only has he helped to maintain the department but has added efficiencies that benefit us all"*. One

of these efficiencies was the Faculty Activity reporting survey and database I created, first for the department and then on behalf of the entire college. This system allowed for ease of faculty submitting their activity to a single location and automated, formatted reporting of faculty activity for efficient monthly and bi-yearly reporting. I met with representatives of both College of Aviation (Kimberly Luthi) and College of Business (Maria Petrescu) to discuss how they could implement a similar system, and the new Watermark activity reporting was partially modeled after my Faculty Reporting system. In addition to serving as Associate Department Chair from February 2023 to May 2025, I was also Acting Department Chair from January 2024 to May 2024 as the college continued to search for a permanent Department Chair. My leadership during this time was both publicly and privately praised, such as John Griffith (former MST chair) stating:

> We have talked before that you would be a great department chair should you want to do that in the future. You really took care of the department in your roles as acting chair and associate chair. You have a bright future. Thanks for being so great to work with!

Beyond the university, I have been an active leader in my overarching professional community, Mathematical Association of America (MAA), as well as the smaller research community Research in Undergraduate Mathematics Education (RUME). As part of the MAA, I have chaired the subcommittee on Technology in Undergraduate Education, served on the MAA Council for Teaching and Learning, and served on the committee for Teaching of Undergraduate Mathematics [**MAA Service Letter**]. As part of RUME, I have served on committees dedicated to leadership nomination and annual conference programming as well as run for leadership positions (Treasurer and Secretary) [**Request to Run for RUME Leadership**]. Beyond my professional communities, I have served several times as a National Science Foundation Grant Reviewer and am a frequent journal reviewer, sought out for my expertise in APOS Theory, proof, and structural equation modeling.

Promise for Sustained Future Development and Scholarly Activity

I plan to continue developing along my three major lines of inquiry outlined in my [**Summary of Scholarly Activity**]: Generative AI in Education, Educational Activities in Online STEM Education, and Data Analysis for Engineering Education.

Regarding work on Generative AI in Education, I have an under review paper that utilizes the automated diagnostic item generation from my prior work diagnosing student misconceptions to assess the efficacy of Generative AI diagnosing potential student thinking. I am also lead editor of a special issue on Generative AI in the Teaching and Learning of Mathematics with *Problems, Resources, and Issues in Mathematics Undergraduate Studies* that is on track to be published by end of year 2027. These efforts are in addition to the 2026-2027 NSF grant *Convening to Investigate the Impact of AI-Powered Pedagogies in Postsecondary Mathematics Education* I am co-PI on alongside MAA's Chief Strategy & Operations Officer Audrey Malagon. These efforts illustrate my ability to weave teaching, scholarly activity, and service together in a manner consistent with my research agenda for productive future activity.

Regarding work on Educational Activities in Online STEM Education, I will continue to apply to NSF IUSE grants in collaboration with Georgia State University to develop diagnostic calculus assessments. Grant submissions in both 2024 and 2025 were reviewed as "Competitive" but were not funding, suggesting the project is meritorious and provides a strong foundation for continued refinement and resubmission [**EG__09 & EG__12 Reviews**]. Note Georgia State University is an R1 university and this collaborative effort would further bolster Embry-Riddle's continuing climb as a research university.

Regarding work on Data Analysis for Engineering Education, I will continue to support research efforts hosted by the ERAU-WW College of Aviation. This allows me to apply my educational data science research expertise directly into ERAU signature research fields. My work in the NSF-funded Research Scholars Program led by Robert Deters (COA) contributed to my inclusion on the under review NSF IUSE proposal led by Ghazal Barai (COA) titled *Appraising Algorithmic Evidence: How AI-Generated Workforce Signals Shape STEM Learners' Motivation and Faculty Reliance on AI.*